

The Fermat-Type Dodge Ladder: Primes $n^{2^k} + 1$ on the $6N$ Skeleton and the Non-Monotonic Richness*

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June 8, 2026

Abstract

Part XXXII read prime richness as the product of per-prime “dodges”—primes forbidden to divide a sequence enhance it. We push that idea into higher degree along the Fermat-type family $n^{2^k} + 1$: $n^2 + 1$, $n^4 + 1$, $n^8 + 1$. The arithmetic is governed by a single cyclotomic fact: $x^{2^k} \equiv -1 \pmod{q}$ is solvable exactly when $q \equiv 1 \pmod{2^{k+1}}$, with 2^k roots. So $n^{2^k} + 1$ can be divided *only* by $q = 2$ and primes $q \equiv 1 \pmod{2^{k+1}}$, a vanishing fraction 2^{-k} of all primes, while at those primes it is hit hard ($\omega = 2^k$). Two findings, both measured by direct (Miller–Rabin / BPSW) primality testing to $n \leq 10^6$ ($n^4 + 1$ up to 10^{24}). (i) The mod-6 *wing* structure is degree-invariant: every $n^{2^k} + 1$ splits its primes right:left = 1:2 exactly as $n^2 + 1$ does, because $n^{2^k} \equiv n^2 \pmod{6}$ for all $k \geq 1$; the exponent is invisible to the coarse lattice. (ii) The Bateman–Horn richness $C = \prod_q g(q)$ is *non-monotonic* in k : as the degree doubles, the dodge widens (more enhancing primes) but the hits deepen, and the two effects balance to a peak at $n^4 + 1$. We measure $C = 1.376, 2.676, 2.075$ for $k = 1, 2, 3$ —rising then falling—with the local product reproducing each to $\lesssim 1\%$, including a degree-eight Bateman–Horn check rarely attempted. The infinitude of primes $n^4 + 1$ is, like Landau’s problem, open; we make no claim about it. As throughout, this is a measurement, not a theorem.

1 Introduction

The richness principle of Part XXXII [7] was simple: a polynomial is prime-rich in proportion to how many small primes its discriminant forbids from dividing it. For $n^2 + 1$ (Part XXXI [6]) the forbidding rule was the character of -1 : only primes $q \equiv 1 \pmod{4}$ can divide, and the rest enhance. Raising the degree along $n^4 + 1$, $n^8 + 1$, and generally $n^{2^k} + 1$ sharpens the rule into a cyclotomic one and lets us watch the dodge tighten.

The governing fact is classical. The congruence $x^{2^k} \equiv -1 \pmod{q}$ has a solution iff the order of -1 , namely 2, is a 2^k -th power in $(\mathbb{Z}/q)^\times$, which happens iff $2^{k+1} \mid q - 1$; equivalently q splits completely in $\mathbb{Q}(\zeta_{2^{k+1}})$. Then there are exactly 2^k roots. So $n^{2^k} + 1$ is divisible only by $q = 2$ and by primes $q \equiv 1 \pmod{2^{k+1}}$, which are a fraction 2^{-k} of all primes by Dirichlet. The sequence dodges an ever-larger share of the primes as k grows—but where it is hit, it is hit on 2^k residues at once.

*Code and measured data: <https://github.com/Ruqing1963/6N-dodge-ladder>

The family has a famous diagonal. At $n = 2$ it is the Fermat numbers $F_k = 2^{2^k} + 1$, the sequence whose first five terms (3, 5, 17, 257, 65537) led Fermat to guess they were all prime—until F_5 was found divisible by 641. The dodge ladder reframes that story: the early F_k sit in a corridor whose escaped-prime share grows like $1 - 2^{-k}$, so small terms meet very few possible divisors and look prime; but the same exponent that widens the dodge sets the depth of any hit at $\omega = 2^k$, and when a divisor finally lands ($641 \mid F_5$) it lands hard. Fermat saw the dodge; he did not see that the penalty for being hit scales with it.

None of this is new as theory, and the infinitude of primes of the form $n^4 + 1$ is as open as Landau's $n^2 + 1$. What we contribute is the $6N$ reading of the ladder, the observation that the wing split is degree-invariant, and a direct high-precision measurement of the resulting richness—which turns out to rise and then fall, peaking at the quartic.

2 Setup: the cyclotomic dodge

For $f(n) = n^{2^k} + 1$ and a prime q let $\omega(q)$ be the number of roots of $f \equiv 0 \pmod{q}$. By the fact above,

$$\omega(q) = \begin{cases} 1, & q = 2, \\ 2^k, & q \equiv 1 \pmod{2^{k+1}}, \\ 0, & \text{otherwise.} \end{cases}$$

The Bateman–Horn richness is $C = \prod_q g(q)$ with $g(q) = (1 - \omega(q)/q)/(1 - 1/q)$, and $Q(N) := \#\{n \leq N : f(n) \text{ prime}\} \sim C \int_2^N dt / \log(t^{2^k} + 1)$. The local factor reads

$$g(q) = \frac{q}{q-1} > 1 \quad (q \text{ dodged}), \quad g(q) = \frac{q-2^k}{q-1} < 1 \quad (q \equiv 1 \pmod{2^{k+1}}, q \text{ hit}).$$

Because the values $n^{2^k} + 1$ grow like n^{2^k} , a root sieve would need primes to N^{2^k} , out of reach; we therefore enumerate by direct primality testing—deterministic Miller–Rabin (twelve bases, valid below 3.3×10^{24} , covering $n^4 + 1$ to $n \leq 10^6$) for $k \leq 2$, and the Baillie–PSW test for $n^8 + 1$. The conditional product for C converges by the Dirichlet balance of the dodged ($g > 1$) against the hit ($g < 1$) primes, exactly as for $n^2 + 1$.

3 The dodge tightens; the wings do not move

Table 1 and Figure 1(A) show the tightening. For $n^2 + 1$ the divisor primes are $q \equiv 1 \pmod{4}$ (5, 13, 17, 29, ...); for $n^4 + 1$ only $q \equiv 1 \pmod{8}$ (17, 41, 73, 89, ...), so 5, 13, 29, 37 are now dodged; for $n^8 + 1$ only $q \equiv 1 \pmod{16}$ (17, 97, 113, ...), dodging 41, 73, 89 as well. Each step halves the share of primes that can divide.

The coarse geometry is untouched. Since squaring fixes the residues $\{0, 1, 3, 4\}$ of $n^2 \pmod{6}$, one has $n^{2^k} \equiv n^2 \pmod{6}$ for every $k \geq 1$, so $n^{2^k} + 1 \pmod{6}$ is identical to $n^2 + 1 \pmod{6}$: odd n are annihilated, $6 \mid n$ sends the value to the right break-zone, $n \equiv 2, 4$ to the left.

Proposition 1. The wing split of $n^{2^k} + 1$ is 1:2 (right:left) for every $k \geq 1$, independent of the degree.

We measure right:left = 0.500, 0.499, 0.517 for $k = 1, 2, 3$ (Fig. 2(B)); the exponent is invisible to the $6N$ lattice. The character that the degree *does* control lives modulo 2^{k+1} , not modulo 6: Figure 2(A) shows the mod-8 spectrum of $n^4 + 1$, with three-quarters of the primes enhancing and the $q \equiv 1 \pmod{8}$ quarter suppressed by $(q-4)/(q-1)$.

f	divides iff	ω	$N_{\max}$	Q	right:left	richness C (meas. / prod.)
$n^2 + 1$	$q \equiv 1 \pmod{4}$	2	10^6	54,109	0.500	1.376 / 1.373
$n^4 + 1$	$q \equiv 1 \pmod{8}$	4	10^6	52,610	0.499	2.676 / 2.679
$n^8 + 1$	$q \equiv 1 \pmod{16}$	8	10^5	2,498	0.517	2.075 / 2.093

Table 1: The dodge ladder. Each doubling of degree restricts the divisor primes to a finer residue class and deepens the hit ($\omega = 2^k$). The wing split is degree-invariant; the richness C rises then falls. (The $n^8 + 1$ sample is smaller, so its measured C carries a $\sim 2\%$ statistical error.)

4 The non-monotonic richness

The richness is a tug-of-war. Widening the dodge adds enhancing factors $q/(q-1) > 1$ over a larger share of primes, pushing C up; deepening the hit replaces the modest $n^2 + 1$ suppression $(q-2)/(q-1)$ with the severe $(q-2^k)/(q-1)$ on the surviving divisor primes, pulling C down. The measured constants (Table 1, Fig. 1(B)) show the balance tipping: C climbs from 1.376 at $n^2 + 1$ to 2.676 at $n^4 + 1$, then *falls* to 2.075 at $n^8 + 1$. The peak at the quartic is not a product artifact: the direct count gives $C_{\text{emp}}(n^4 + 1) = 2.676$ against $C_{\text{emp}}(n^8 + 1) = 2.075$, a 22% drop far exceeding the 2% sampling error. Throughout, the local product reproduces the measured count to better than a percent—a Bateman–Horn verification carried, for $n^8 + 1$, to degree eight, where values reach 10^{40} and direct primality testing is the only route. The two ends of the ladder make the balance plain. As $k \rightarrow \infty$ the share of primes that can divide, 2^{-k} , tends to zero—almost every prime is dodged, an unbounded supply of enhancements—yet at the vanishing minority of hit primes the value is split 2^k ways at once, a rarer and rarer but ever deeper destruction. Neither extreme maximises C : the unbounded-dodge limit is throttled by the unbounded-depth hits, and the two trends meet their best compromise at $k = 2$, where the richness peaks.

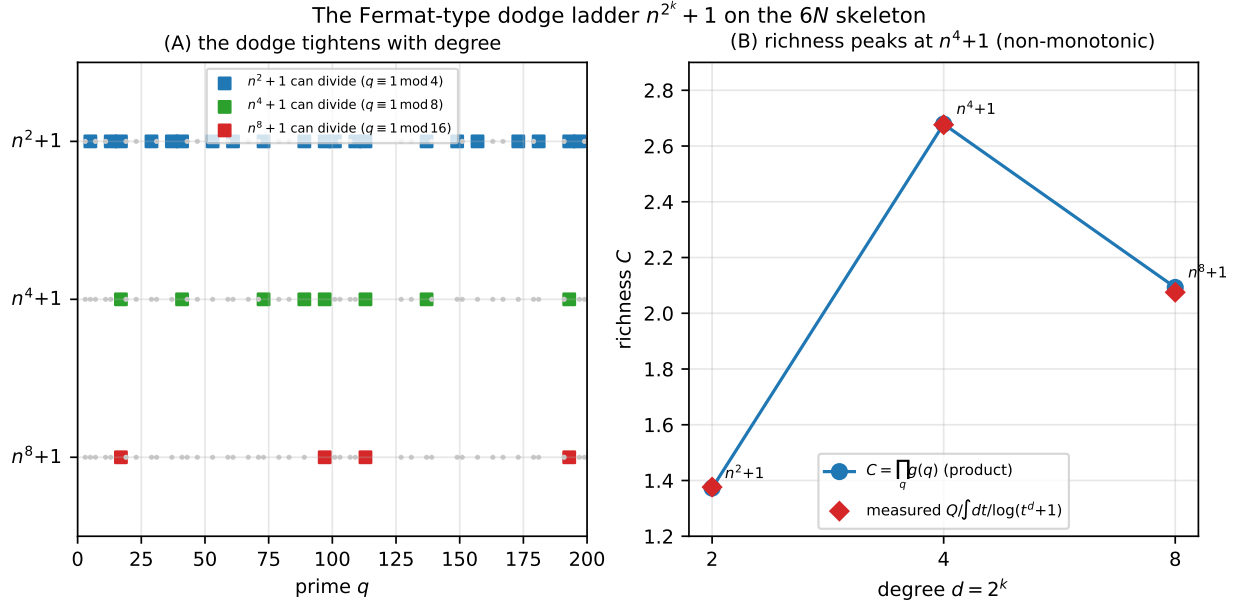


Figure 1: (A) The dodge map: the primes that can divide $n^{2^k} + 1$ (coloured) thin out as the degree doubles, restricted to $q \equiv 1 \pmod{2^{k+1}}$; dodged primes in grey. (B) The richness C versus degree: the product (blue) and the measured count (red) agree, and both peak at $n^4 + 1$ before falling at $n^8 + 1$.

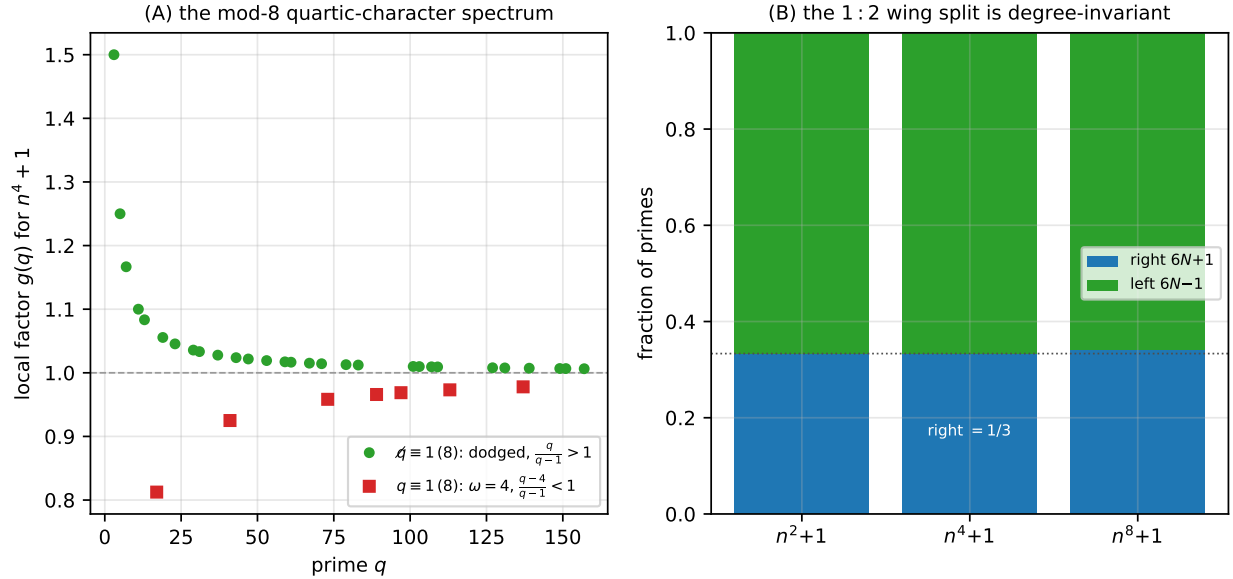


Figure 2: (A) The mod-8 quartic-character spectrum of $n^4 + 1$: three-quarters of primes are dodged ($g > 1$), the $q \equiv 1 \pmod{8}$ quarter suppressed with $\omega = 4$. (B) The 1:2 wing split (right = $1/3$) is identical for $n^2 + 1$, $n^4 + 1$, $n^8 + 1$: the $6N$ mask does not see the exponent.

5 Scope and honest ledger

The Fermat-type ladder shows one mechanism resolved into two moving parts. The exponent sets a cyclotomic dodge— $n^{2^k} + 1$ is divisible only by $q \equiv 1 \pmod{2^{k+1}}$, a fraction 2^{-k} of primes, with $\omega = 2^k$ where it is hit—and the resulting richness is non-monotonic, peaking at $n^4 + 1$ because a wider dodge buys enhancement while a deeper hit costs suppression. The coarse $6N$ geometry, by contrast, is blind to the exponent: the wing split stays pinned at 1:2.

We are explicit about the boundary. The solvability of $x^{2^k} \equiv -1 \pmod{q}$ and its root count are classical cyclotomic facts; the constant C and the Bateman–Horn asymptotic are standard, and the $n^2 + 1$ constant is Landau–Shanks. What this paper adds is the $6N$ reading of the ladder, the degree-invariance of the wings, and a direct high-precision measurement of the richness—including the non-monotonic peak and a degree-eight check. We make no claim about the infinitude of primes $n^{2^k} + 1$ for any $k \geq 1$: that is open already for $k = 1$ (Landau) and no finite computation can settle it; the counts we report would look the same whatever the truth. Everything here equals a direct primality count, takes the Hardy–Littlewood heuristic as input, and asserts no infinitude.

Data and code availability. The Miller–Rabin / BPSW enumeration for $n^2 + 1$, $n^4 + 1$, $n^8 + 1$, the local-factor products, the dodge and $\omega(q)$ tables, and the figure scripts are openly available at <https://github.com/Ruqing1963/6N-dodge-ladder>.

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